

White Paper 1 — Assembling the ranked DEG target list

Pipeline step: Assemble ranked DEG target list **Inputs:** `meta_de_PPvsNN.csv` (meta-analysis differential expression), pathway/regulon evidence **Outputs:** `target_universe.csv`, `fig_target_volcano_annotated.png`

1. Purpose

The starting point of any target-discovery pipeline is a defensible list of disease-associated genes. In psoriasis, the relevant contrast is **lesional (PP) versus non-lesional (NN) skin**: genes whose expression changes between the two states are candidate drivers of the psoriatic phenotype. This white paper documents how we converted a raw meta-analysis differential-expression (DE) table into a ranked, evidence-weighted target universe of 2,379 significant genes, and why each analytical choice was made.

2. Differential expression as the primary signal

2.1 Why RNA-seq differential expression

Psoriasis is a disease of dysregulated transcription — keratinocyte hyperproliferation and a type-17 immune infiltrate both leave strong transcriptomic footprints. The reference psoriasis transcriptome study of Li and colleagues established that lesional skin carries thousands of differentially expressed genes relative to uninvolved skin, and that these genes are enriched for immune and epidermal-differentiation programs [1]. That result is the biological justification for treating the PP-vs-NN DE table as the primary evidence layer.

2.2 Data provenance: recount3 and GENCODE

Uniform quantification matters when combining studies. The upstream expression matrices were drawn from **recount3**, which reprocesses tens of thousands of human RNA-seq samples through a single pipeline so that counts are comparable across studies [2]. Gene models follow the **GENCODE** annotation, the reference human gene set that defines the transcript boundaries and gene identifiers used throughout [3,4].

2.3 The linear-modeling framework: limma-voom

Per-study DE was computed with the **limma** framework, which fits gene-wise linear models and applies empirical-Bayes moderation of the variance — borrowing information across genes to stabilize the per-gene error estimate, which is decisive when per-group sample sizes are modest [5]. Because RNA-seq data are counts with a strong mean–variance relationship, raw counts violate the constant-variance assumption of linear models. The **voom** transformation solves this by estimating the mean–variance trend and converting it into precision weights on log-counts-per-million, so that ordinary linear modeling becomes valid on sequencing data [6]. The limma-voom pairing is the workhorse of the per-study analysis.

3. Combining studies: random-effects meta-analysis

Individual psoriasis cohorts differ in platform, depth, and patient composition, so effect sizes are not identical across studies — they are drawn from a distribution. We therefore combined per-study log-fold-changes with a **random-effects meta-analysis** using the DerSimonian–Laird estimator of between-study variance (τ^2), which explicitly models heterogeneity rather than assuming a single true effect [7]. Each gene's meta-analytic summary carries a combined effect (logFC), its standard error, a z-statistic, and a heterogeneity index (I^2). For STAT3, the meta-analysis returned logFC = +1.25 (up in lesional skin), z = 5.54, and I^2 = 91%, the last indicating substantial but directionally-consistent cross-study heterogeneity.

4. Multiple-testing control

Testing ~23,000 genes simultaneously guarantees false positives at any fixed per-test α . We controlled the **false discovery rate (FDR)** with the Benjamini–Hochberg procedure, which bounds the expected proportion of false discoveries among the genes called significant and is the standard for genome-wide expression screens [8]. Combined with an effect-size floor, the significance rule was **FDR < 0.05 and |log2FC| > 1**, yielding **2,379 significant meta-DEGs** (996 up, 1,383 down).

5. Layering pathway and regulatory evidence

A gene that is both differentially expressed *and* central to a psoriasis-relevant pathway is a stronger candidate than one that is merely differentially expressed. Two orthogonal evidence layers were added:

- **Pathway leading-edge membership.** Gene-set enrichment analysis (GSEA) identifies coordinately regulated gene sets and reports the "leading-edge" subset that drives each enrichment ^[9]. Genes in the leading edge of psoriasis-relevant Hallmark sets (e.g. IL6-JAK-STAT3 signaling, TNF α -via-NF- κ B, interferon- γ response) were flagged.
- **Transcription-factor regulon membership.** TF-activity inference with the **CollectRI** regulon collection ^[10], run through the **decoupleR** statistical framework ^[11], identifies targets of key transcription factors. STAT3's regulon (435 targets) was used to flag genes under its transcriptional control.

6. The composite evidence score

The four evidence layers were combined into a single ranking statistic:

```
evidence_score = 0.35 z(|logFC|) + 0.30 z(-log10 FDR)
                + 0.15 z(robustness) + 0.20 z(pathway_support)
```

where `robustness = k (1 - I2/100)` rewards genes replicated across studies (k) with low heterogeneity, and each term is z-scored across the significant set. The weighting privileges effect size and statistical confidence while giving meaningful credit to cross-study robustness and pathway centrality. The result is `target_universe.csv`, the ranked list that all downstream steps consume.

6b. From ranked list to interaction map

The ranked target universe is a list, and a list treats genes as independent. Because disease phenotypes emerge from the coordinated action of interacting proteins ^[12], the significant DEG set is also passed to a protein–protein interaction network analysis (see the companion explainer, White Paper 1b) that identifies the network-central "hub" proteins organizing the psoriatic response — a step that, in this project, independently placed STAT3 at the topological crossroads of the disease wiring.

7. Deliverables

- `target_universe.csv` — 2,379 significant genes with per-gene meta-statistics, evidence layers, and the composite score.
- `fig_target_volcano_annotated.png` — volcano plot (meta-analytic logFC vs -log10 FDR) with the lead targets and key psoriasis genes labeled.

8. Caveats

- Meta-analytic effect sizes for high- I^2 genes (STAT3 among them) should be read as directionally robust but quantitatively uncertain.
- The evidence-score weights are a defensible but not unique choice; the underlying per-gene statistics are provided so the ranking can be recomputed under alternative weightings.

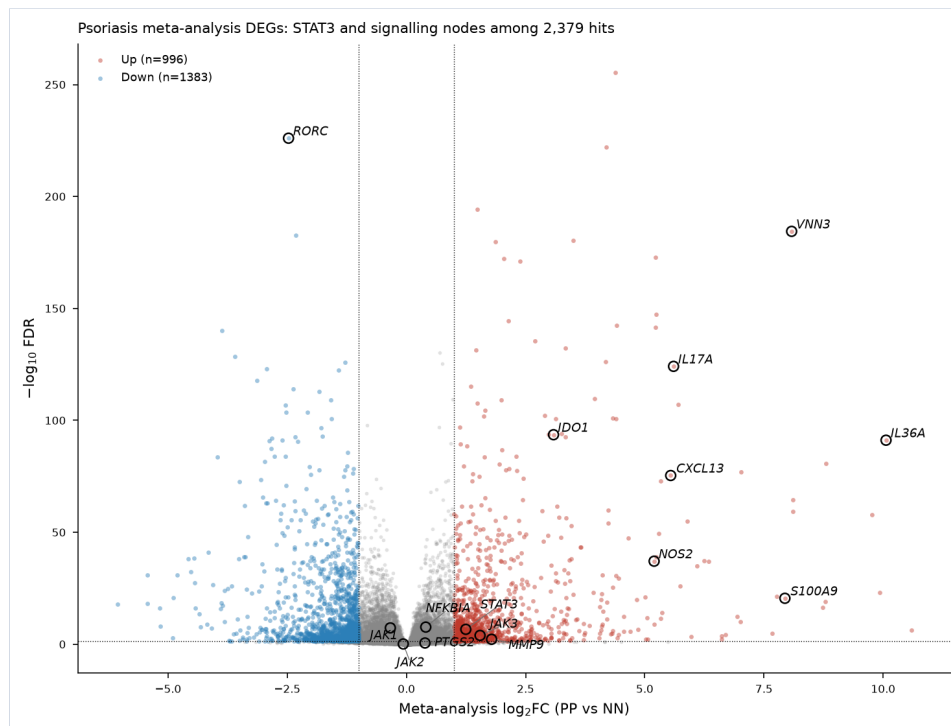


Figure: key output of this pipeline step.

References

1. Li et al. Transcriptome Analysis of Psoriasis in a Large Case–Control Sample: RNA-Seq Provides Insights into Disease Mechanisms. *Journal of Investigative Dermatology*. 2014. doi:[10.1038/jid.2014.28](https://doi.org/10.1038/jid.2014.28)
2. Wilks et al. recount3: summaries and queries for large-scale RNA-seq expression and splicing. *Genome Biology*. 2021. doi:[10.1186/s13059-021-02533-6](https://doi.org/10.1186/s13059-021-02533-6)
3. Harrow et al. GENCODE: The reference human genome annotation for The ENCODE Project. *Genome Research*. 2012. doi:[10.1101/gr.135350.111](https://doi.org/10.1101/gr.135350.111)
4. Frankish et al. GENCODE reference annotation for the human and mouse genomes. *Nucleic Acids Research*. 2018. doi:[10.1093/nar/gky955](https://doi.org/10.1093/nar/gky955)
5. Ritchie et al. limma powers differential expression analyses for RNA-sequencing and microarray studies. *Nucleic Acids Research*. 2015. doi:[10.1093/nar/gkv007](https://doi.org/10.1093/nar/gkv007)
6. Law et al. voom: precision weights unlock linear model analysis tools for RNA-seq read counts. *Genome Biology*. 2014. doi:[10.1186/gb-2014-15-2-r29](https://doi.org/10.1186/gb-2014-15-2-r29)
7. DerSimonian et al. Meta-analysis in clinical trials. *Controlled Clinical Trials*. 1986. doi:[10.1016/0197-2456\(86\)90046-2](https://doi.org/10.1016/0197-2456(86)90046-2)
8. Benjamini et al. Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing. *Journal of the Royal Statistical Society Series B: Statistical Methodology*. 1995. doi:[10.1111/j.2517-6161.1995.tb02031.x](https://doi.org/10.1111/j.2517-6161.1995.tb02031.x)
9. Subramanian et al. Gene set enrichment analysis: A knowledge-based approach for interpreting genome-wide expression profiles. *Proceedings of the National Academy of Sciences*. 2005. doi:[10.1073/pnas.0506580102](https://doi.org/10.1073/pnas.0506580102)
10. Müller-Dott et al. Expanding the coverage of regulons from high-confidence prior knowledge for accurate estimation of transcription factor activities. *Nucleic Acids Research*. 2023. doi:[10.1093/nar/gkad841](https://doi.org/10.1093/nar/gkad841)
11. Badia-i-Mompel et al. decoupleR: ensemble of computational methods to infer biological activities from omics data. *Bioinformatics Advances*. 2022. doi:[10.1093/bioadv/vbac016](https://doi.org/10.1093/bioadv/vbac016)
12. Barabási et al. Network medicine: a network-based approach to human disease. *Nature Reviews Genetics*. 2010. doi:[10.1038/nrg2918](https://doi.org/10.1038/nrg2918)